Practical 1: Create a Maven Project
To Create a maven project we can create file structure manually or via “mvn” command-line

To create manually we need to make a file structure like this:

```
my-app1/
│
├── pom.xml
│
├── src/
│   ├── main/
│   │   ├── java/
│   │   │   └── com/
│   │   │       └── example/
│   │   │           └── myapp1/
│   │   │               └── App.java
│   │   │
│   │   ├── resources/
│   │   │   └── application.properties
│   │   │
│   │   └── webapp/            (only for web projects)
│   │       ├── WEB-INF/
│   │       │   └── web.xml
│   │       └── index.jsp
│   │
│   └── test/
│       ├── java/
│       │   └── com/
│       │       └── example/
│       │           └── myapp1/
│       │               └── AppTest.java
│       │
│       └── resources/
│
└── target/   (generated after build)
```

Or We can use command:
mvn archetype:generate -DgroupId=com.tech -DartifactId=my-app1
-DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

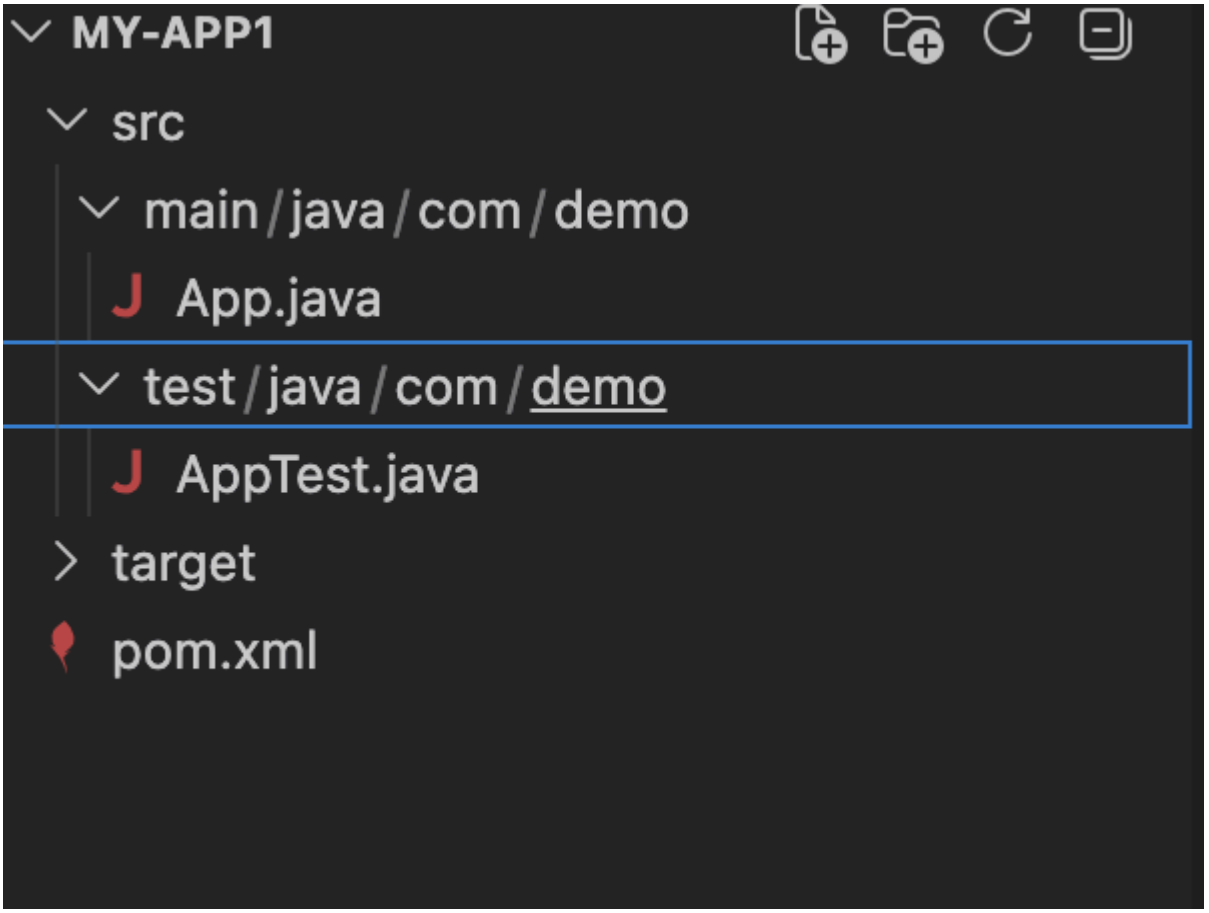
Explaintion:
1. mvn archetype:generate
mvn: This is the Maven command-line tool.
archetype:generate: This tells Maven to generate a project using a Maven archetype.
An archetype is a template for creating projects with a predefined structure
2. -DgroupId=com.example
-DgroupId: This specifies the group ID for the project.
com.example: The value of the groupId. This typically represents the organization or

company creating the project (e.g., a domain name in reverse order).

3.-DartifactId=my-app1
-DartifactId: This specifies the artifact ID for the project.
my-app1: The value of the artifactId. This is usually the name of your project and will be the base name of the generated .jar or .war file.
4. -DarchetypeArtifactId=maven-archetype-quickstart
-DarchetypeArtifactId: Specifies which Maven archetype to use for project creation.
maven-archetype-quickstart: The value of the archetypeArtifactId. This is a commonly used archetype that creates a simple Java project with a basic directory structure and a sample class.
5. -DinteractiveMode=false
-DinteractiveMode: Determines whether Maven should prompt you for input during the process.
false: Disables interactive mode, meaning Maven uses the specified options (groupId, artifactId, etc.) without prompting.
This command is used to quickly bootstrap a new Java project with Maven, giving you a starting
point for development.

```
sumitkumarmehta@sumits-MacBook-Air lemaven % mvn archetype:generate -DgroupId=com.demo -DartifactId=my-app1 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 50 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 66 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml (1.1 kB at 32 kB/s)
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 59 MB/s)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /Users/sumitkumarmehta/lemaven
[INFO] Parameter: package, Value: com.demo
[INFO] Parameter: groupId, Value: com.demo
[INFO] Parameter: artifactId, Value: my-app1
[INFO] Parameter: packageName, Value: com.demo
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /Users/sumitkumarmehta/lemaven/my-app1
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  1.165 s
[INFO] Finished at: 2026-04-19T18:27:05+05:30
[INFO] ----------------------------------------------------------------------------
sumitkumarmehta@sumits-MacBook-Air lemaven %
```

Project Built Successfull



As we can see the file structure it created
Now let's try to run hello world

A hello world code is pre-written in src->main->java->com here we can find App.java

mvn compile → it compiles main code
Note: It does not run test cases.
It only compile .java → .class and stores it in the target folder.

```
sumitkumarmehta@sumits-MacBook-Air my-app1 % mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< com.demo:my-app1 >------------------------
[INFO] Building my-app1 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ my-app1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/sumitkumarmehta/lemaven/my-app1/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ my-app1 ---
[INFO] Recompiling the module because of added or removed source files.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target/classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.345 s
[INFO] Finished at: 2026-04-19T18:33:45+05:30
[INFO] ------------------------------------------------------------------------
sumitkumarmehta@sumits-MacBook-Air my-app1 %
```

mvn test → Compile + run test

```
sumitkumarmehta@sumits-MacBook-Air my-app1 % mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< com.demo:my-app1 >------------------------
[INFO] Building my-app1 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ my-app1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/sumitkumarmehta/lemaven/my-app1/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ my-app1 ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ my-app1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/sumitkumarmehta/lemaven/my-app1/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ my-app1 ---
[INFO] Recompiling the module because of added or removed source files.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target/test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
  not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ my-app1 ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit.JUnit3Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.demo.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.010 s -- in com.demo.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.976 s
[INFO] Finished at: 2026-04-19T18:34:37+05:30
[INFO] ------------------------------------------------------------------------
sumitkumarmehta@sumits-MacBook-Air my-app1 %
```

mvn package → compile + run test + create jar/war and store it in “my-app1/target/my-app1-1.0-SNAPSHOT.jar”

Now we can run this using java -cp target/my-app1-1.0-SNAPSHOT.jar com.demo.App

```
sumitkumarmehta@sumits-MacBook-Air my-app1 % java -cp target/my-app1-1.0-SNAPSHOT.jar com.demo.App
Hello World!
sumitkumarmehta@sumits-MacBook-Air my-app1 %
```